

PROJECT DOCUMENTATION

Import data using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-4228

1. INTRODUCTION

1.1 Project Overview

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. Bulk employee data is received in Excel format and migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is configured to map source fields to the appropriate target table fields.

Upon executing the transformation, records are created in the target table and validated to ensure data integrity. The project demonstrates the end-to-end data import process, including data loading, field mapping, transformation execution, and result verification.

1.2 Project Purpose

- Understand the complete spreadsheet-to-ServiceNow import workflow.
- Use an Import Set table as the staging area for incoming employee data.
- Map source fields to target fields using a Transform Map.
- Validate the imported records after transformation.

2. IDEATION PHASE

2.1 Problem Statement

Manual migration of spreadsheet records can be time-consuming and may cause data entry errors, duplicate records, and inconsistent information. A structured import process is therefore required to move employee data into ServiceNow accurately.

2.2 Initial Preparation

The project begins by preparing a Google Spreadsheet with sample employee data and downloading it in Sample Spreadsheet.xlsx format. The spreadsheet acts as the external source from which the employee records are loaded into ServiceNow.

2.3 Custom Table Preparation

A custom ServiceNow table named Employee Test is created to store the transformed employee records. The table name is **u_employee_test**. The required fields are created as String fields:

- Employee ID – String
- Employee Name – String
- Email – String
- Department – String
- Location – String

After creating all required fields, the form is saved and the fields are verified on the Employee Test form.

3. REQUIREMENT ANALYSIS

3.1 Import Set Requirement

The prepared spreadsheet is loaded through Load Data. An Import Set table named Employee Import is created, with the name **u_employee_import**. This table acts as the staging area before the records are transformed.

3.2 Transform Map Requirement

A Transform Map named Sample Spreadsheet Import is created. The source table is Employee Import and the target table is Employee Test. Auto Map Matching Fields and Mapping Assist are used to connect the source and target fields.

3.3 Data Transformation Requirement

- The Transform Map must map the employee fields correctly.
- The transformation must create records in the Employee Test table.
- The process must provide a success state after transformation.
- Imported records must be verified in the target table.

3.4 Duplicate Prevention Requirement

Coalesce is enabled on a field map to avoid duplicate employee records. When the same employee identifier is imported again, the existing record is updated instead of creating another record.

3.5 Validation Requirement

- The transform state should show Completed and Success when processing finishes.
- Transform History should show inserted, updated, skipped, or ignored rows.
- The Employee Test table should reflect the final transformed records.

3.6 Reporting Requirements

- Employees by Department – Pie Chart, grouped by Department, Count aggregation.
- Employees by Location – Bar Chart, grouped by Location, Count aggregation.
- Employee List Report – List report showing Employee ID, Employee Name, Email, Department and Location.

3.7 Dashboard Requirement

The three reports are added to an Employee Analytics Dashboards dashboard so that employee information can be viewed centrally. The dashboard provides a single place to view the reports and monitor the imported employee data.

4. PROJECT DESIGN

4.1 Proposed Workflow

The solution follows a simple sequence: prepare the spreadsheet → create the Employee Test target table → create the Employee Import staging table → create the Transform Map → map source and target fields → execute the transformation → verify records → enable Coalesce for repeated imports → create reports → add the reports to a dashboard.

4.2 Data Flow

Spreadsheet (Excel) → Import Set Table (Employee Import) → Transform Map (Sample Spreadsheet Import) → Employee Test Target Table → Reports and Dashboard.

4.3 Detailed Transformation Logic

The spreadsheet is uploaded through Load Data and stored in the Import Set staging table. The Transform Map then maps source fields to the corresponding Employee Test fields. When Transform is executed, the

records are created in the target table. Coalesce is later enabled so repeated imports can update matching records rather than inserting duplicates.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

1. Phase 1: Prepare the spreadsheet with sample employee records.
2. Phase 2: Create the Employee Test custom table and required fields.
3. Phase 3: Create the Employee Import staging table.
4. Phase 4: Create the Sample Spreadsheet Import Transform Map and map fields.
5. Phase 5: Execute the transformation and validate the Employee Test table.
6. Phase 6: Enable Coalesce and test duplicate prevention.
7. Phase 7: Insert additional Excel data and inspect transform history.
8. Phase 8: Create employee reports.
9. Phase 9: Add reports to the Employee Analytics Dashboards dashboard.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Testing Performed

- The initial employee spreadsheet was imported successfully and records were created in Employee Test.
- Repeated employee IDs with changed names or email values were tested with Coalesce.
- A second upload containing four rows resulted in two inserted new IDs and two updated existing IDs.
- Uploading the same sheet again resulted in four ignored rows, with zero new insertions and zero updates.

7. RESULTS

7.1 Result Overview

The following pages present the actual project workflow and screenshots from the supplied project document. The results show the spreadsheet preparation, ServiceNow table creation, Import Set loading, Transform Map configuration, field mapping, transformation execution, duplicate prevention, report creation, and dashboard setup.

7.2 Verification Summary

- Employee records are loaded into the Employee Import staging table before transformation.
- The Transform Map connects the source Employee Import fields with the target Employee Test fields.
- Coalesce prevents duplicate records during repeated imports and allows matching records to be updated.
- Reports provide department-wise, location-wise, and detailed employee views.
- The Employee Analytics Dashboards dashboard brings the reports together in one view.

7.3 Output Sections

The screenshots in the remaining results pages document each major operation in sequence, from preparing the spreadsheet through the final dashboard. Each result section uses the actual screenshots supplied with the project content.

7.4 Transition to Screenshot Results

The first result steps are spreadsheet preparation and creation of the Employee Test custom table. These establish the source data and target structure required for the later import and transformation stages.

7.5 Key Result Summary

Data import: The employee spreadsheet is loaded into the Employee Import staging table and transformed into the Employee Test table.

Duplicate handling: With Coalesce enabled, matching employee identifiers are updated instead of creating duplicate records.

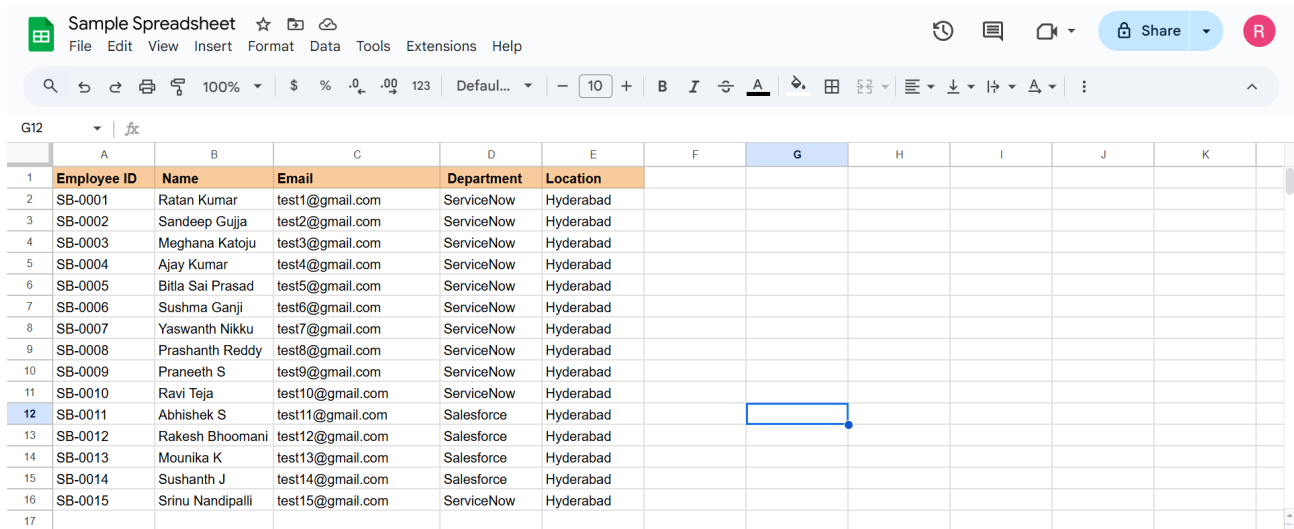
Repeated upload test: One four-row upload produced two inserted records and two updated records. Uploading the same sheet again produced four ignored rows, with zero new insertions and zero updates.

Reporting and dashboard: Employees by Department, Employees by Location, and Employee List reports are created and added to the Employee Analytics Dashboards dashboard.

7. RESULTS

7.1 Prepare the Spreadsheet

A Google Spreadsheet is created with sample employee records and downloaded as an Excel file for the ServiceNow import.

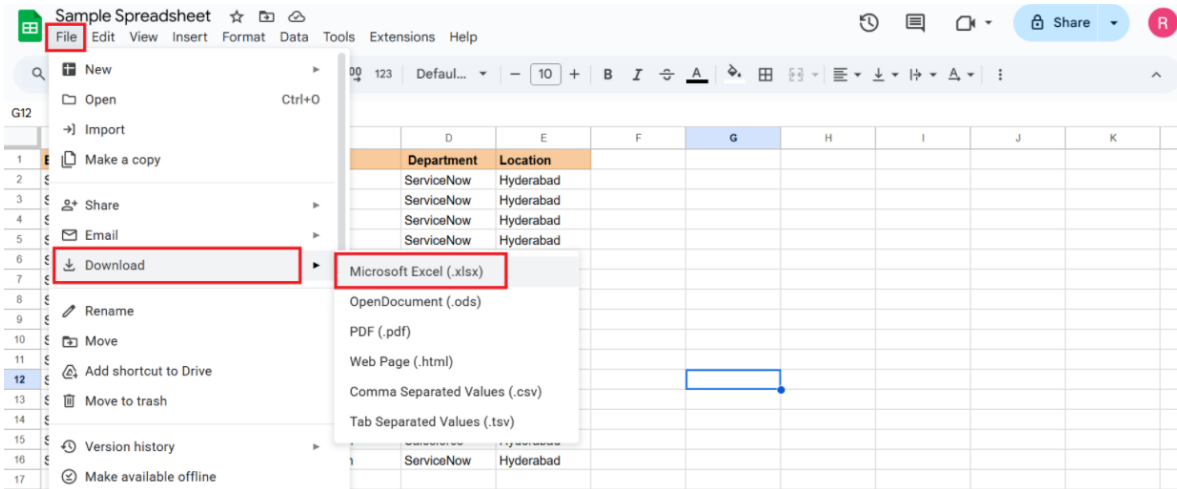


	A	B	C	D	E	F	G	H	I	J	K
1	Employee ID	Name	Email	Department	Location						
2	SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad						
3	SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad						
4	SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad						
5	SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad						
6	SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad						
7	SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad						
8	SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad						
9	SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad						
10	SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad						
11	SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad						
12	SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad						
13	SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad						
14	SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad						
15	SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad						
16	SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad						
17											

Screenshot – Prepare the Spreadsheet

7.2 Create the Employee Test Table

The Employee Test custom table is created and configured with Employee ID, Employee Name, Email, Department and Location fields.

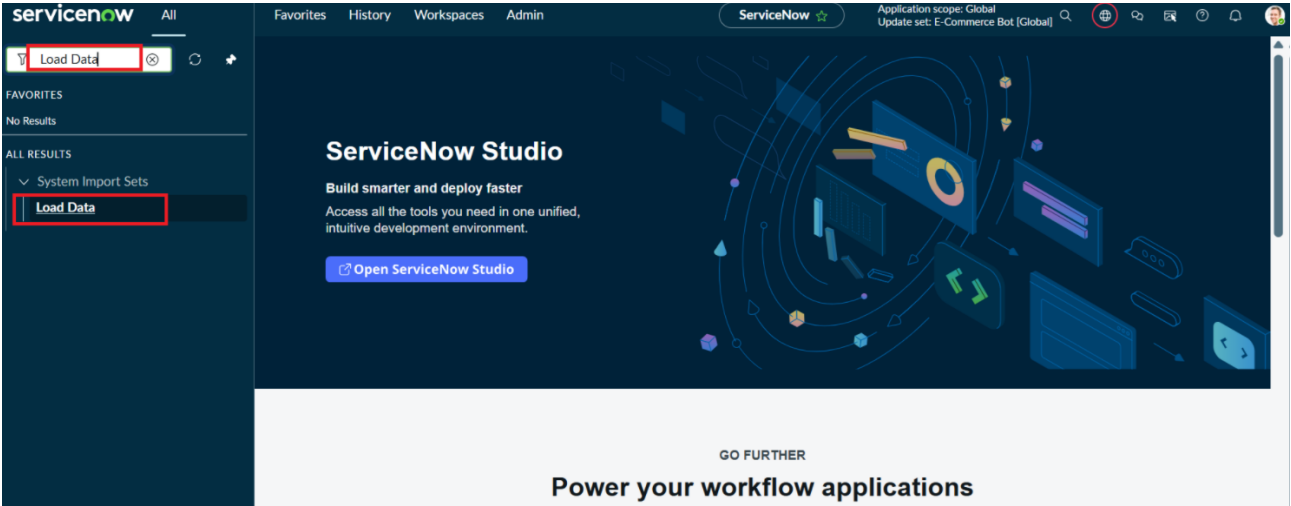


Screenshot – Employee Test Table

7. RESULTS

7.3 Create the Import Set Table

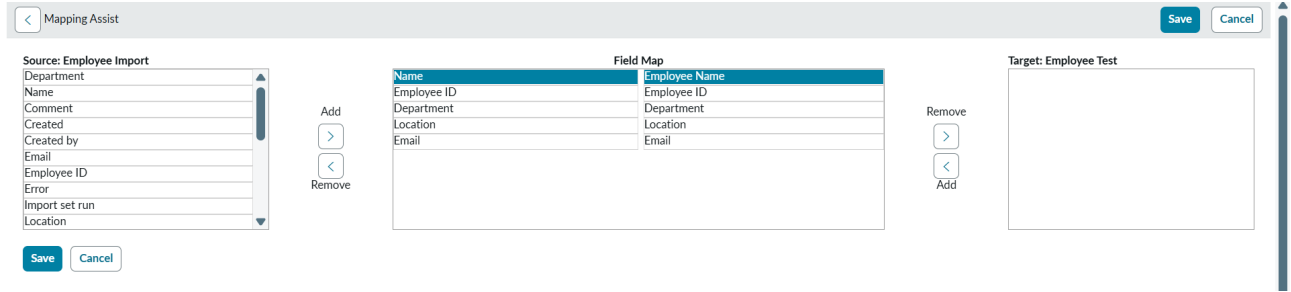
Using Load Data, the Employee Import table is created. The table name is u_employee_import and it is used as the staging area for the spreadsheet data.



Screenshot – Import Set Creation

7.4 Create the Transform Map

The Sample Spreadsheet Import Transform Map is created with Employee Import as the source and Employee Test as the target.



Screenshot – Transform Map Creation

7. RESULTS

7.5 Map the Fields

Auto Map Matching Fields and Mapping Assist are used to map the source fields to the corresponding target fields.

Progress

Name

ImportProcessor

State

Complete

Completion code

Success

Message

Processed: 15, inserts 15, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.368)

Next steps...

Import sets

Go to the import sets for this data load

Loaded data

Go to the newly imported data inside the staging table: u_employee_import

Create transform map

Create a transform map for the newly staged data

Import log

View the import log

Screenshot – Field Mapping

7.6 Execute the Transformation

The Transform button is used to execute the import. The process is checked for a successful completion state.

Sample Spreadsheet Import

Create new record on empty coalesce fields

Copy

Update

Delete

Related Links

[Auto Map Matching Fields](#)

[Mapping Assist](#)

[Validate Coalesce Fields](#)

[Transform](#)

[Index Coalesce Fields](#)

[Run Point Scan](#)

[\[SN Utils\] Versions \(1\)](#)

Field Maps (5)

Transform Scripts

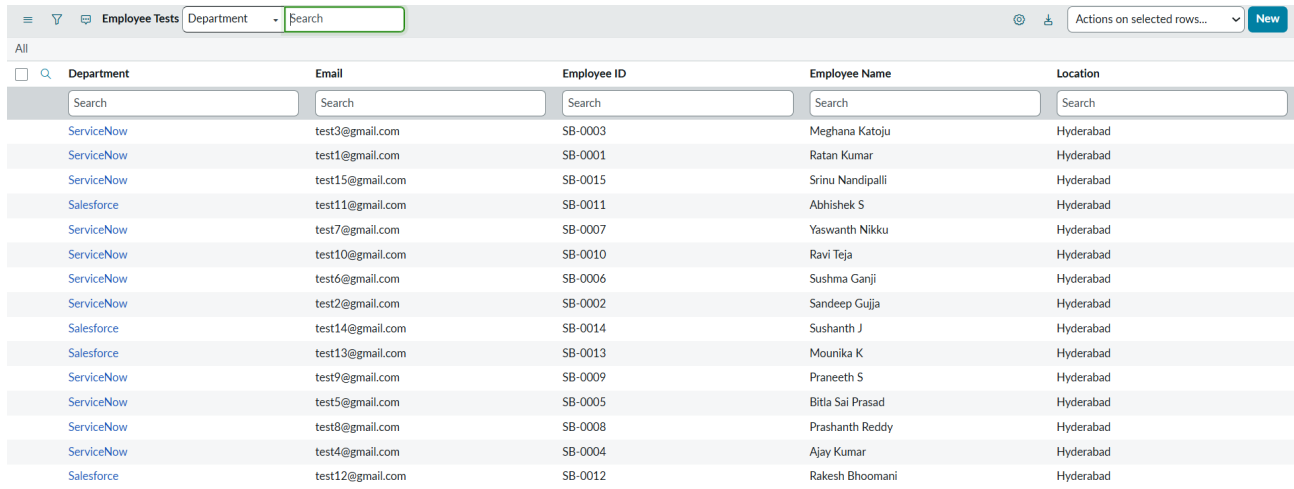
	Source field	Target field	Coalesce
☐	u_name	u_employee_name	false
	u_employee_id	u_employee_id	false
	u_department	u_department	false
	u_location	u_location	false
	u_email	u_email	false

Screenshot – Transform Execution

7. RESULTS

7.7 Verify the Imported Records

After the import succeeds, the Employee Test table contains the transformed employee records. The list can be rearranged using Personalised List Columns for verification.

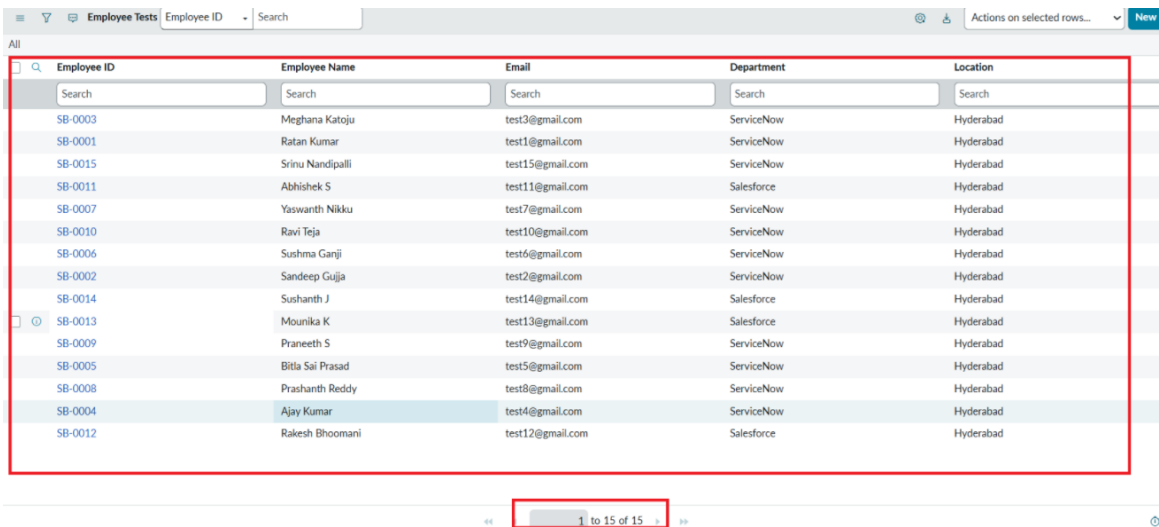


Department	Email	Employee ID	Employee Name	Location
ServiceNow	test3@gmail.com	SB-0003	Meghana Katoju	Hyderabad
ServiceNow	test1@gmail.com	SB-0001	Ratan Kumar	Hyderabad
ServiceNow	test15@gmail.com	SB-0015	Srinu Nandipalli	Hyderabad
Salesforce	test11@gmail.com	SB-0011	Abhishek S	Hyderabad
ServiceNow	test7@gmail.com	SB-0007	Yaswanth Nikku	Hyderabad
ServiceNow	test10@gmail.com	SB-0010	Ravi Teja	Hyderabad
ServiceNow	test6@gmail.com	SB-0006	Sushma Ganji	Hyderabad
ServiceNow	test2@gmail.com	SB-0002	Sandeep Guja	Hyderabad
Salesforce	test14@gmail.com	SB-0014	Sushanth J	Hyderabad
Salesforce	test13@gmail.com	SB-0013	Mounika K	Hyderabad
ServiceNow	test9@gmail.com	SB-0009	Praneeth S	Hyderabad
ServiceNow	test5@gmail.com	SB-0005	Bitla Sai Prasad	Hyderabad
ServiceNow	test8@gmail.com	SB-0008	Prashanth Reddy	Hyderabad
ServiceNow	test4@gmail.com	SB-0004	Ajay Kumar	Hyderabad
Salesforce	test12@gmail.com	SB-0012	Rakesh Bhoomani	Hyderabad

Screenshot – Final Imported Records

7.8 Enable Coalesce

A field in the Transform Map is set to Coalesce = True. This allows existing employee records to be updated when matching data is imported again, while new employee IDs can be inserted.



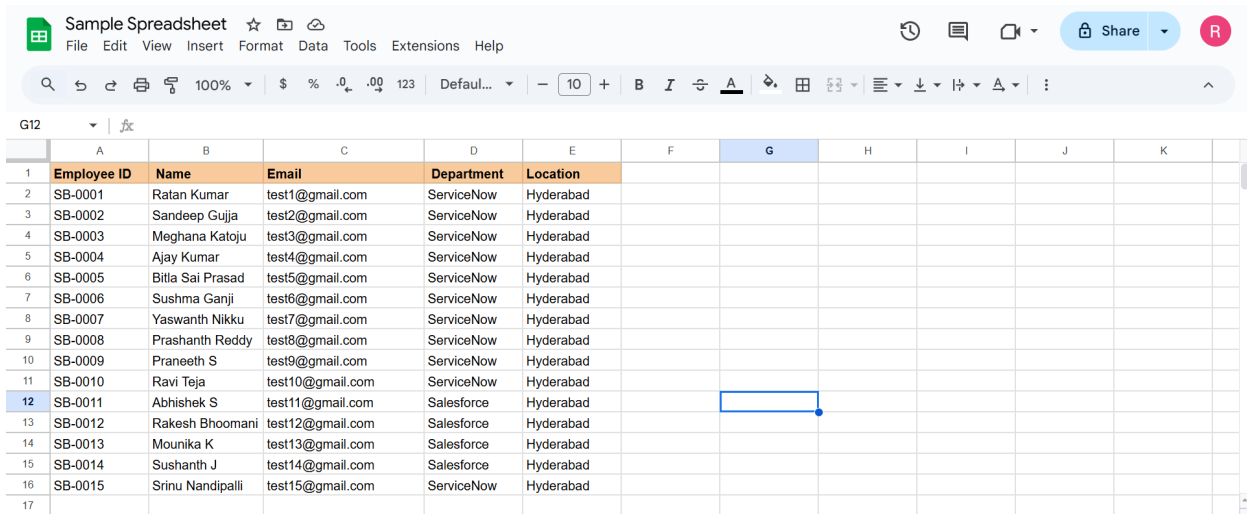
Employee ID	Employee Name	Email	Department	Location
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad

Screenshot – Coalesce Configuration

7. RESULTS

8. Inserting New Data in Excel Format

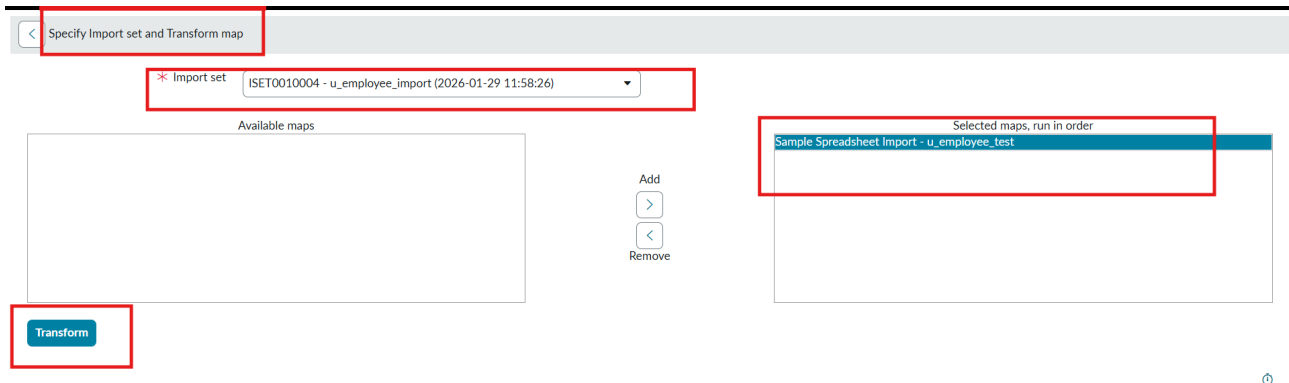
A new Excel file is loaded into the existing Employee Import table. The sheet number and header row are selected before submitting the import.



	A	B	C	D	E	F	G	H	I	J	K
1	Employee ID	Name	Email	Department	Location						
2	SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad						
3	SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad						
4	SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad						
5	SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad						
6	SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad						
7	SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad						
8	SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad						
9	SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad						
10	SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad						
11	SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad						
12	SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad						
13	SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad						
14	SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad						
15	SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad						
16	SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad						
17											

Screenshot – New Excel Data

The new file contains four rows: two rows match existing employee IDs but contain changed values, while two rows contain new employee IDs.



Specify Import set and Transform map

Import set: ISET0010004 - u_employee_import (2026-01-29 11:58:26)

Available maps

Selected maps, run in order

Sample Spreadsheet Import - u_employee_test

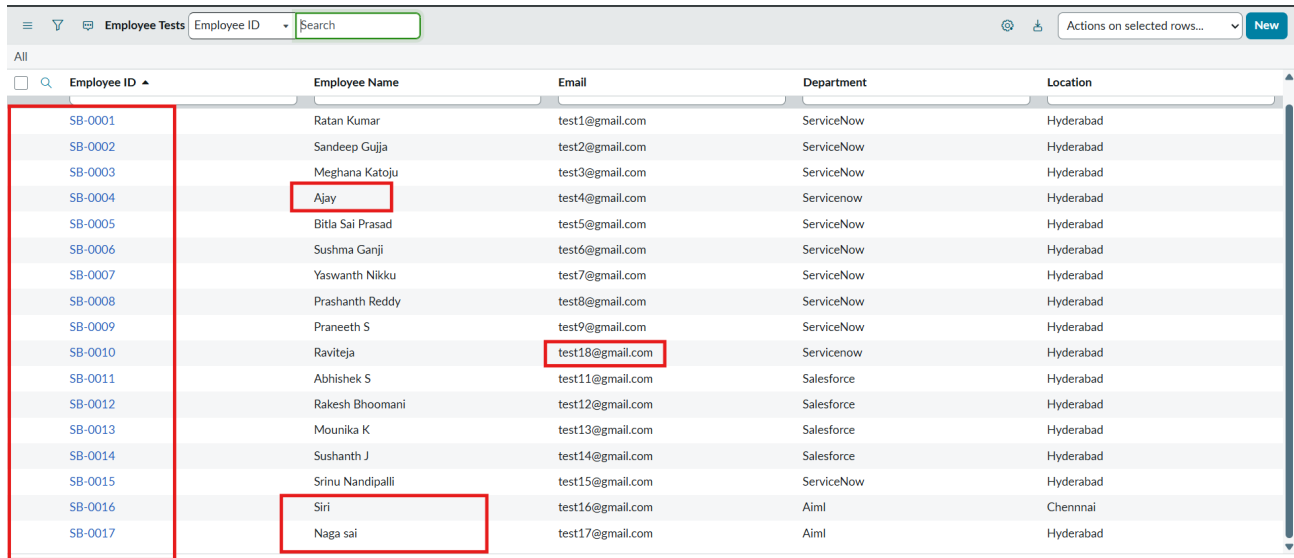
Transform

Screenshot – Transform/Update Result

7. RESULTS

8.1 Transform History – Inserted and Updated Records

After transformation, Transform History shows the result of processing the four uploaded rows. Two new employee IDs are inserted and two existing employee records are updated.



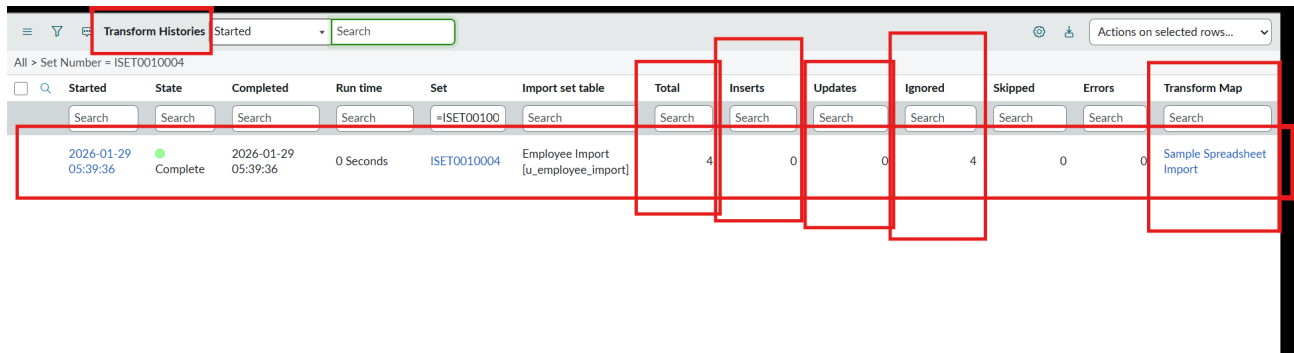
The screenshot displays a table titled 'Employee Tests' with columns: Employee ID, Employee Name, Email, Department, and Location. A red box highlights the first 17 rows (SB-0001 to SB-0017). Within this range, two rows are further highlighted with red boxes: SB-0004 (Employee Name: Ajay) and SB-0010 (Email: test19@gmail.com). The table shows a mix of new records and updates to existing ones.

Employee ID	Employee Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Raviteja	test19@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad
SB-0016	Siri	test16@gmail.com	Aiml	Chennnai
SB-0017	Naga sai	test17@gmail.com	Aiml	Hyderabad

Screenshot – Inserted and Updated Records

8.2 Repeated Import Test

The same Excel sheet is uploaded again using the same Transform Map. The result shows four ignored rows, with zero inserted and zero updated records.



The screenshot displays a table titled 'Transform Histories' with columns: Started, State, Completed, Run time, Set, Import set table, Total, Inserts, Updates, Ignored, Skipped, Errors, and Transform Map. A red box highlights the first row, which shows a completed import for the set 'ISET0010004' with 4 ignored rows and 0 inserts or updates. The 'Transform Map' column for this row is 'Sample Spreadsheet Import'.

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-01-29 05:39:36	Complete	2026-01-29 05:39:36	0 Seconds	ISET0010004	Employee Import [u_employee_import]	4	0	0	4	0	0	Sample Spreadsheet Import

Screenshot – Repeated Import Result

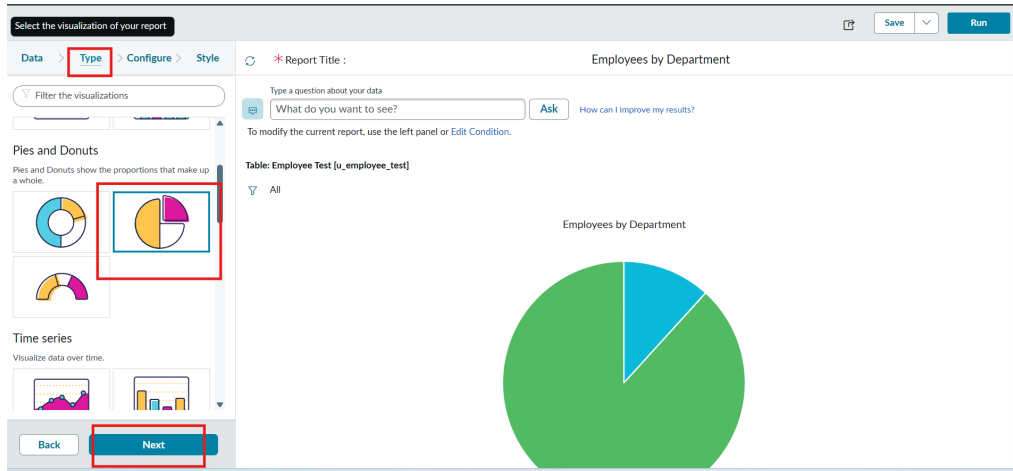
7. RESULTS

9. Create Reports

Three reports are created from the Employee Test table to provide different views of the imported employee data.

9.1 Employees by Department

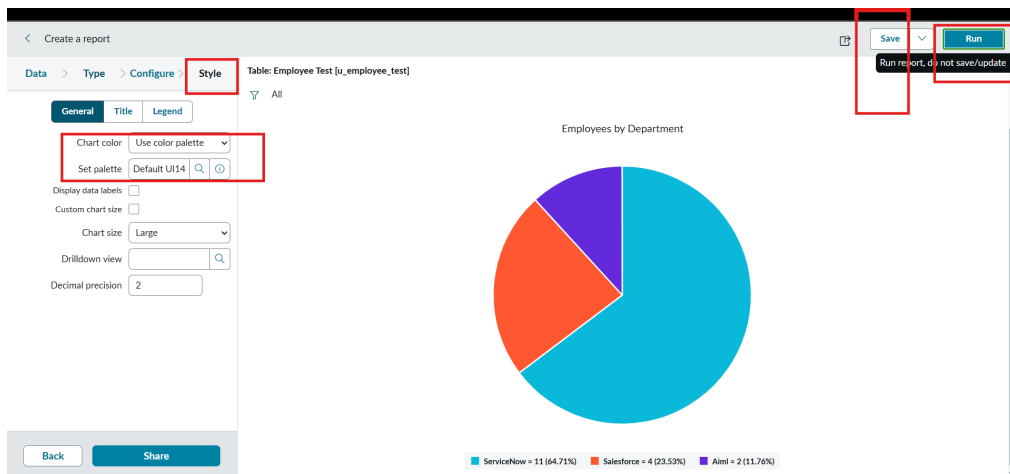
A Pie Chart is configured with Department as the Group By field and Count as the aggregation.



Screenshot – Employees by Department

9.2 Employees by Location

A Bar Chart is configured using Location as the Group By field and Count as the aggregation.



Screenshot – Employees by Location

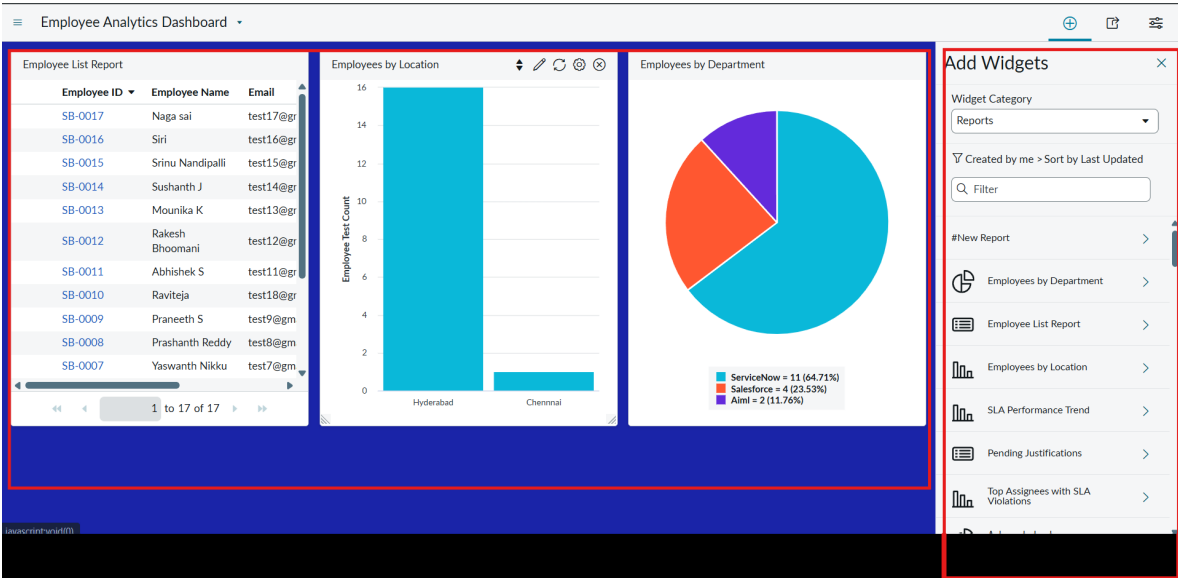
9.3 Employee List Report

A List report displays Employee ID, Employee Name, Email, Department and Location.

7. RESULTS

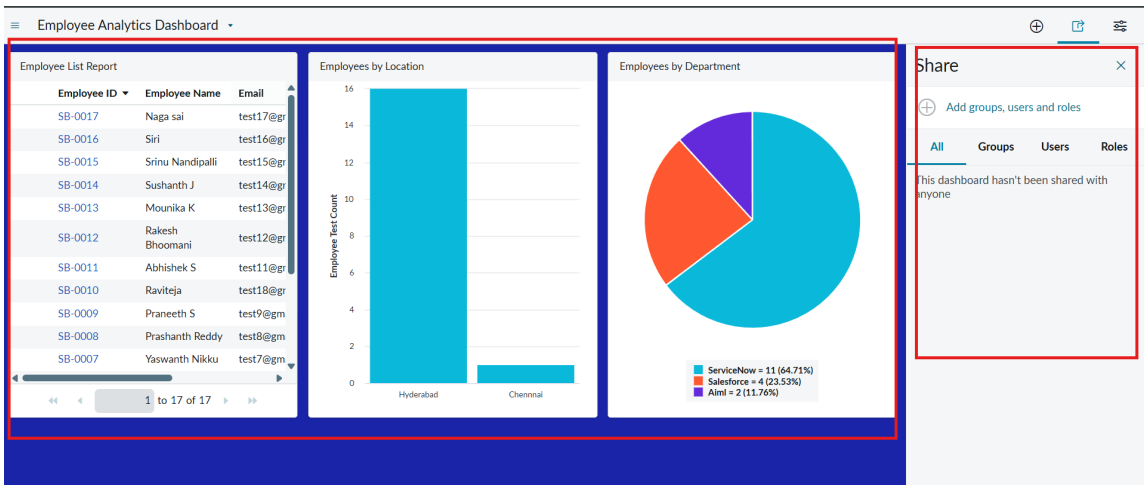
10. Add Reports to Dashboard

An Employee Analytics Dashboards dashboard is created. The Employees by Department, Employees by Location and Employee List Report are added to the dashboard.



Screenshot – Employee Analytics Dashboard

The dashboard provides a centralized visual view of the employee information. Users can also configure dashboard widgets and restrict access to users, groups and roles through the sharing options.



Screenshot – Dashboard with Reports

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Duplicate Prevention: Coalesce on Employee ID prevents repeated employee records and supports updates to existing data.
- Accuracy and Data Integrity: Transform Maps provide structured field mapping between the spreadsheet and target table.
- Visibility: Transform History shows inserted, updated, skipped and ignored results.
- Reporting: Reports and dashboards provide a centralized view of employee information.

8.2 Disadvantages

- The process requires correct spreadsheet structure and field mapping before transformation.
- Incorrect or inconsistent source values may require correction before importing.
- Dashboard and report configuration requires additional ServiceNow setup.

9. CONCLUSION

This project successfully implements an employee data management workflow using ServiceNow Import Sets and Transform Maps. Spreadsheet data is first placed into a staging table, mapped to the Employee Test target table, transformed, and verified.

A key enhancement is the use of Coalesce on the Employee ID field. It enables the system to identify existing employee records during repeated imports, preventing duplicate entries and allowing changed values to update the existing record.

The project is further extended with reports and an Employee Analytics Dashboards dashboard. These provide visual information about employee records, department-wise and location-wise distribution, and list-based employee data, giving administrators a centralized view of the imported information.

Overall, the project demonstrates a practical end-to-end ServiceNow spreadsheet import workflow with transformation, duplicate prevention, validation, reporting and dashboard visualization.